

Inspect a user-process page table ([easy](#))

```
mariu@DESKTOP-0K07MJA: x + v
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C5B pa 0x87F1F000 perm 0x5B
va 0x2000 pte 0x21FC7817 pa 0x87F1E000 perm 0x17
va 0x3000 pte 0x21FC7407 pa 0x87F1D000 perm 0x7
va 0x4000 pte 0x21FC70D7 pa 0x87F1C000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
ugetpid_test starting
usertrap(): unexpected scause 0xd pid=3
```

Speed up system calls:

```
hart 2 starting
hart 1 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC7C5B pa 0x87F1F000 perm 0x5B
va 0x1000 pte 0x21FC705B pa 0x87F1C000 perm 0x5B
va 0x2000 pte 0x21FC6C17 pa 0x87F1B000 perm 0x17
va 0x3000 pte 0x21FC6807 pa 0x87F1A000 perm 0x7
va 0x4000 pte 0x21FC64D7 pa 0x87F19000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFD000 pte 0x21FD4813 pa 0x87F52000 perm 0x13
va 0xFFFFE000 pte 0x21FD08C7 pa 0x87F40000 perm 0xC7
va 0xFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
ugetpid_test starting
ugetpid_test: OK
print_kpgtbl starting
```

Print page table:

```
marium@DESKTOP-0K07MJA: X + v
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .
^$/d' > user/dorphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-
inter -ggdb -gdwarf-2 -DSOL_PGTBL -DLAB_PGTBL -MD -mcmodel=medany -ffreesta
no-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-
fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-lo
ultin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-
putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -
tin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie
user/pgtbltest.o user/pgtbltest.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_pgtbl
r/pgtbltest.o user/_lib.o user/_sys.o user/_printf.o user/_umalloc.o
riscv64-unknown-elf-objdump -S user/_pgtbltest > user/pgtbltest.asm
riscv64-unknown-elf-objdump -t user/_pgtbltest | sed '1,/SYMBOL TABLE/d; s/
/^$/d' > user/pgtbltest.sym
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user
user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs u
rtests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user
n user/_pgtbltest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blo
total 2000
ballocc: first 984 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M
nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none
raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
page table 0x0000000086f51000
..0x0000000000000000: pte 0x0000000021bd3401 pa 0x0000000086f4d000
.. ..0x0000000000000000: pte 0x0000000021bd3001 pa 0x0000000086f4c000
.. .. ..0x0000000000000000: pte 0x0000000021bd381b pa 0x0000000086f4e000
.. .. ..0x0000000000000000: pte 0x0000000021bd2c17 pa 0x0000000086f4b000
.. .. ..0x0000000000000000: pte 0x0000000021bd2807 pa 0x0000000086f4a000
.. .. ..0x0000000000000000: pte 0x0000000021bd2417 pa 0x0000000086f49000
..0x000000003fc0000000: pte 0x0000000021bd4001 pa 0x0000000086f50000
.. ..0x000000003ffffe0000: pte 0x0000000021bd3c01 pa 0x0000000086f4f000
.. .. ..0x000000003fffffd000: pte 0x0000000021bd4813 pa 0x0000000086f52000
.. .. ..0x000000003fffffe000: pte 0x0000000021bd5807 pa 0x0000000086f56000
.. .. ..0x000000003ffffff000: pte 0x000000002000180b pa 0x0000000080006000
init: starting sh
$ |
```

```
marium@DESKTOP-0K07MJA: X + v
.. .. ..0x0000000000002000: pte 0x0000000021bd2807 pa 0x0
.. .. ..0x0000000000003000: pte 0x0000000021bd2417 pa 0x0
..0x000000003fc0000000: pte 0x0000000021bd4001 pa 0x00000000
.. ..0x000000003ffffe0000: pte 0x0000000021bd3c01 pa 0x0000
.. .. ..0x000000003fffffd000: pte 0x0000000021bd4813 pa 0x0
.. .. ..0x000000003fffffe000: pte 0x0000000021bd5807 pa 0x0
.. .. ..0x000000003ffffff000: pte 0x000000002000180b pa 0x0
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21BC7C5B pa 0x86F1F000 perm 0x5B
va 0x1000 pte 0x21BC705B pa 0x86F1C000 perm 0x5B
va 0x2000 pte 0x21BC6C17 pa 0x86F1B000 perm 0x17
va 0x3000 pte 0x21BC6807 pa 0x86F1A000 perm 0x7
va 0x4000 pte 0x21BC64D7 pa 0x86F19000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x21BD1413 pa 0x86F45000 perm 0x13
va 0x3FFFFFFE000 pte 0x21BD00C7 pa 0x86F40000 perm 0xC7
va 0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
ugetpid_test starting
```

Make grade:

```
== Test answers-pgtbl.txt ==
answers-pgtbl.txt: OK
== Test usertests ==
$ make qemu-gdb
Timeout! (300.3s)
== Test time ==
time: OK
Score: 28/41
make: *** [Makefile:370: grade] Error 1
marium@DESKTOP-0K07MJA:~/xv6-labs-2025$ |
```